

Mohammed Alam

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EDUCATION

Columbia University – *Bachelor of Science in Computer Science*

Expected May 2027

Relevant Coursework: Data Structures & Algorithms, Computer Systems, Advanced Programming (C), Database Systems, Web Security, User Interface Design, Software Quality & Design (Java)

EXPERIENCE

Columbia Housing

May 2024 – Present

Senior Systems & Operations

New York, NY

- Engineered Node.js services and REST APIs to automate workflows across 40 properties and 2M+ sq. ft.
- Developed React and Next.js tools to manage housing data across 6,000+ rooms, improving staff visibility into room status
- Integrated AWS S3, Asana REST API, and Google Apps Script, cutting manual work 80% across recurring 6,000+ tasks

Headstarter AI

July 2024 – September 2024

Software Engineering Fellow

New York, NY

- Delivered 5 applications in 8 weeks using Next.js, TypeScript, OpenAI, Pinecone, and Stripe
- Won 1st place among 75+ teams at a TikTok-sponsored hackathon by building a real-time React blogging platform
- Enabled semantic search across 10,000+ documents through RAG pipelines using OpenAI and Pinecone

Freelance

September 2019 – May 2024

Software Engineer

New York, NY

- Created 500+ websites, Discord bots, and automations using React, Node.js, MongoDB, and Redis
- Optimized client apps by caching frequently accessed MongoDB data in Redis
- Established structured logging, error tracking, and alerting to accelerate incident resolution

PROJECTS

Mini-Vercel

July 2026 – Present

- Architected self-hosted CI/CD converting GitHub pushes into HTTPS deployments with TypeScript and Docker
- Achieved zero-downtime releases by health-checking containers and atomically switching Nginx routes on AWS EC2
- Routed project subdomains through a Go reverse proxy with deployment controls via CLI and Next.js dashboard

Guess The Number

October 2021 – June 2024

- Scaled a multiplayer Discord game to 13M+ members across 13K+ servers through organic adoption
- Reduced command latency by 60% by sharding Redis and MongoDB data across live game sessions
- Maintained sub-200ms p95 response times during traffic spikes with Bull.js background job queues

Dorm Room Viewer | Columbia Housing

June 2025 – Present

- Secured Columbia Housing adoption with a React explorer for room layouts, sizes, photos, and walk times
- Served floor plans and photos across 40 buildings through an AWS S3 image pipeline with fallback handling
- Connected Columbia CAS SSO through NextAuth.js with authenticated admin tools for managing housing listings

Doorman | Columbia Housing

August 2026 – Present

- Launched a TypeScript, React, and Express Asana builder now used in Columbia Housing's daily operations
- Extended Asana's API with custom bulk operations, project editing, and workflow controls
- Automated spreadsheet imports, validation, and batched project synchronization with live build progress

SKILLS

Languages: JavaScript, TypeScript, Python, Java, Go, SQL

Technologies: React, Next.js, React Native, Node.js, PostgreSQL, MongoDB, Redis, Supabase, AWS, Docker, Nginx, Git, REST APIs